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COSC 5580
Fall 2023
Project Assignment 3 - Report

Hypothesis Testing

Hypothesis-1

The null hypothesis, H_0 is that *there is no significant difference between means of the two features - **CPI US** and **CPI JP*** and the alternative hypothesis, H_1 is that *there is significant different between the means of the two features*. The t-statistic, corresponding p-value and degrees of freedom is $statistic = 100.14, pvalue = 0.0, df = 21774.0$. We are using 0.05 as the significance threshold. Since, the pvalue is less than 0.05, we can reject the null hypothesis and accept the alternative hypothesis. therefore, we can conclude that there is a significant difference in the CPI values of USA and Japan in our dataset.

Hypothesis-2

We used a t-test because we have two groups (countries with high and low government debt), and want to compare their means. The t-test is appropriate when dealing with a continuous outcome variable and comparing the means of two independent groups. First, we should choose the data we need from our data set, which is exchange rate data for countries classified into two groups based on government debt levels. Then, we need to define groups. Group 1 is the countries with high government debt, and another group is countries with low government debt. With our hypotheses, we need to perform a t-test and conduct an independent samples t-test to compare the means of government debt between the two groups. To interpret results, If the p-value is less than the set significance level (often 0.05), the null hypothesis is rejected. This shows sufficient data to establish that exchange rates differ significantly between nations with high and low government debt levels. The t-test is appropriate in this situation because it allows people to determine if the observed variations in exchange rates between the two groups are statistically significant or if they may have happened by coincidence. It is critical to ensure that the t-test assumptions are met, particularly the assumptions of normality and homogeneity of variances between groups. The t-test value were - $t - statistics$: [104.00] and $p - value$: [0.].

Classification

Support Vector Machine (SVM)

Details of the model used

- Best Hyperparameters: 'C': 10, 'kernel': 'linear'
- Accuracy: 0.78
- ROC AUC: 0.94
- Classifier Threshold Point - False Positive Rate: 0.11
- True Positive Rate: 0.81
- Equal Opportunity Score: 0.08

Interpretation. Using a 10 fold cross validation. Based on the statistics generated for our predictive task data, the model prefers a linear decision boundary with a moderate regularization rate. 78% of the test set was predicted correctly. The ROC score is pretty good (good discrimination), which means that our model does not perform based on random chance and it is better, the closer our ROC value to 1 is, the better. The equal opportunity score looks great and means that the model is pretty fair. Overall, the accuracy of the model could be improved by performing more testing and experimenting with hyperparameters and changing the cross validation rate (10 fold validation is pretty good), but currently experimenting with so many hyperparameters creates a lot of load on the system.

Decision Tree

Details of the model used

- Best Hyperparameters: 'criterion': 'entropy', 'max_depth': None, 'min_samples_split': 5
- Accuracy: 0.65
- ROC AUC: 0.66
- Classifier Threshold Point - False Positive Rate: 0.41
- True Positive Rate: 0.71
- Equal Opportunity Score: 0.12

Interpretation. The decision tree model does slightly better than the random forest model. The model uses entropy criterion, which creates a large number of small partitions. The ROC score indicated that the model is not taking a random guess, but the value is not too great and not anywhere close to the ideal value (which is 1). The equal opportunity score for this model is better than Random forest, which indicates that for our data a single tree's prediction is more fair, than the random forests' multiple tree. (Even though RF works on a majority wins basis, sometimes there are biases between trees and some trees decisions are given higher weights, so this could be one of the possible reasons).

Random Forests

Details of the model used

- Best Hyperparameters: 'criterion': 'entropy', 'max_depth': None, 'n_estimators': 200
- Accuracy: 0.62
- ROC AUC: 0.66
- Classifier Threshold Point - False Positive Rate: 0.49
- True Positive Rate: 0.72
- Equal Opportunity Score: 0.21

Interpretation. Random forest uses gini impurity by default, but by using grid search, entropy was better suited for our data which creates a large number of small partitions. 200 decision trees were created by the RF model, but even other options like 50 or 100 trees performed similarly, when I individually checked, however the model determined 200 as the best hyperparameter. 62% of the test cases were predicted correctly. The ROC score is not that great, and it is pretty close to the random guess line, so the model has a limited discrimination power. The Equal opportunity indicates that the model is biased to some extent but it is not very far away from 0 (which is supposed to be the best score).

Logistic Regression

- Best Hyper-parameters : 'C': 10
- Accuracy: 0.93
- Equal Opportunity Difference : 0.04

Interpretation. 'C' is the inverse of regularization strength. A 'C' value of 10 indicates that the model benefits from a higher complexity to capture the patterns in the dataset effectively. Logistic regression gives the highest accuracy, of 93%. An EOD value of 0.04 indicates low bias in model's predictions.

Naive Bayes

- Accuracy: 0.57
- Equal Opportunity Difference : -0.66

Interpretation. Naive Bayes didn't have any hyper parameters that we could search over. The accuracy is considerably lower as compared to Logistic Regression. The negative value of the EOD suggests a high level of bias in the model's predictions.

KNN

- Best Hyper-parameters : 'n-neighbors': 3
- Accuracy: 0.60
- Equal Opportunity Difference : -0.56

Interpretation. Similar to KNN, the negative value of EOD of this magnitude indicates a considerable bias in favor of one group over the other. The rest of the results are self explanatory.

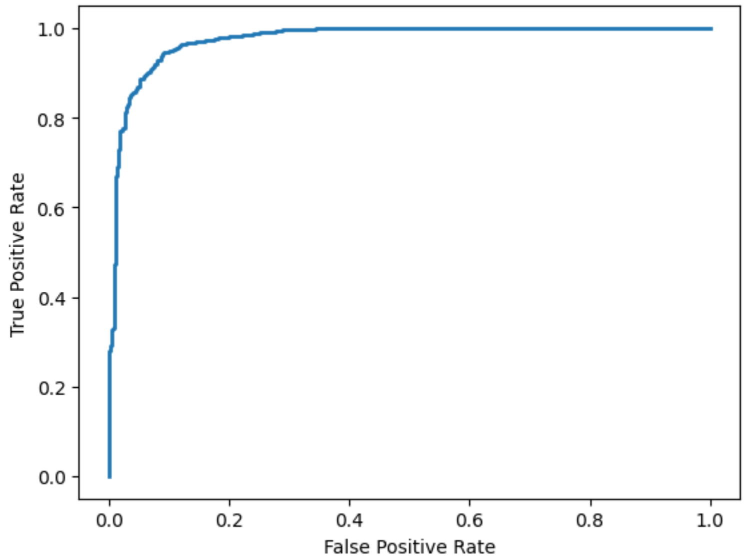

Figure 1: Logistic Regression ROC Curve

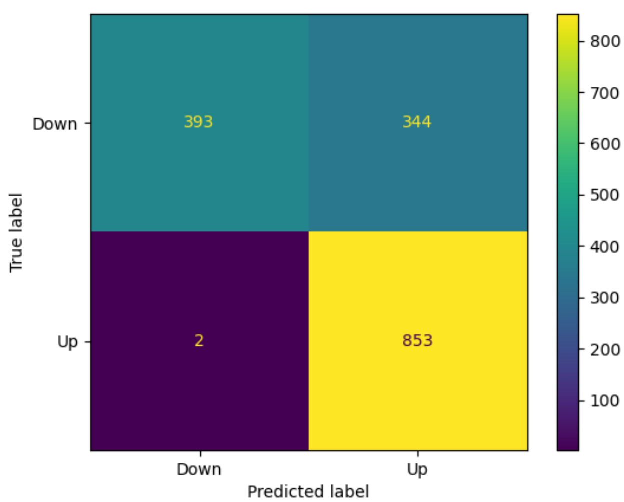

Figure 2: SVM Confusion Matrix

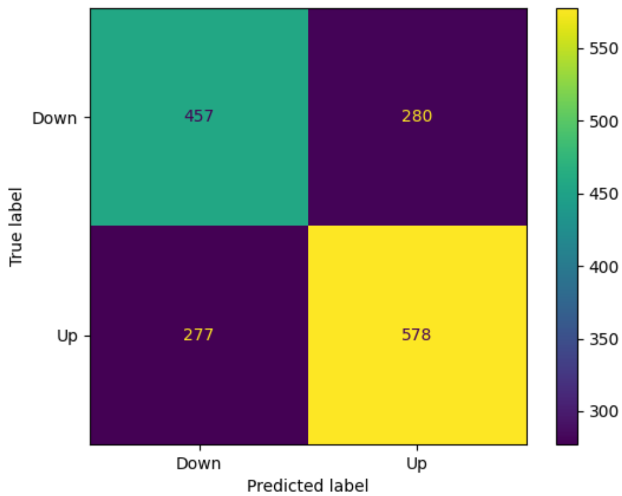

Figure 3: Decision Tree Confusion Matrix

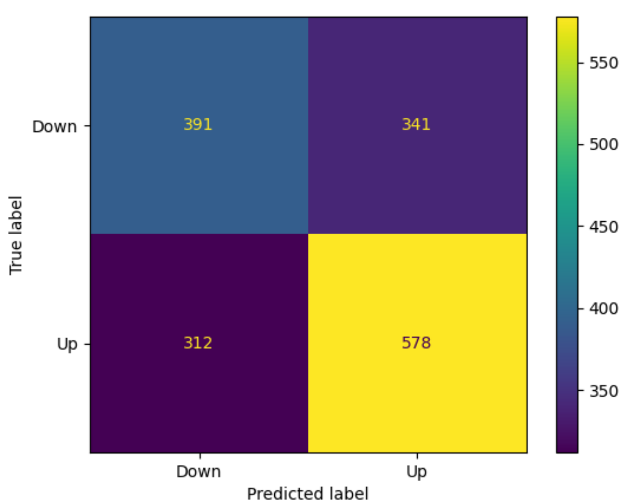

Figure 4: KNN Confusion Matrix

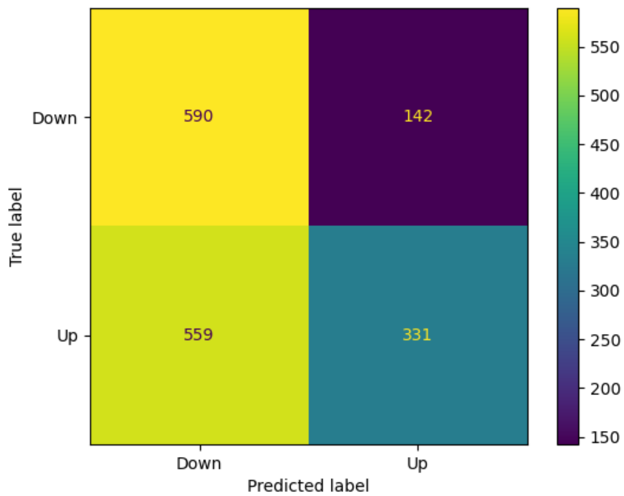

Figure 5: Naive Bayes Confusion Matrix

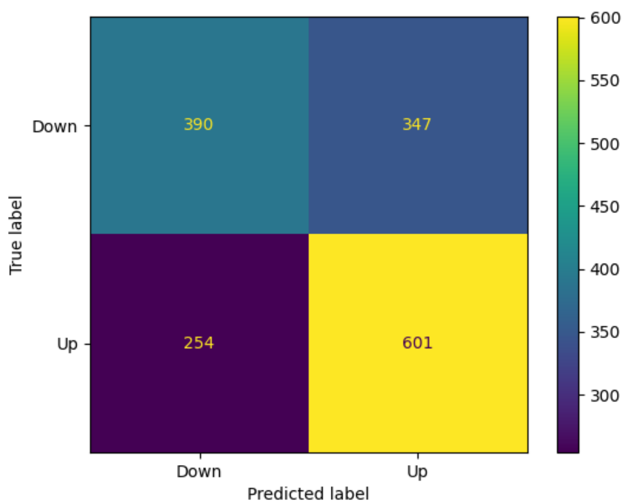

Figure 6: Random Forest Confusion Matrix

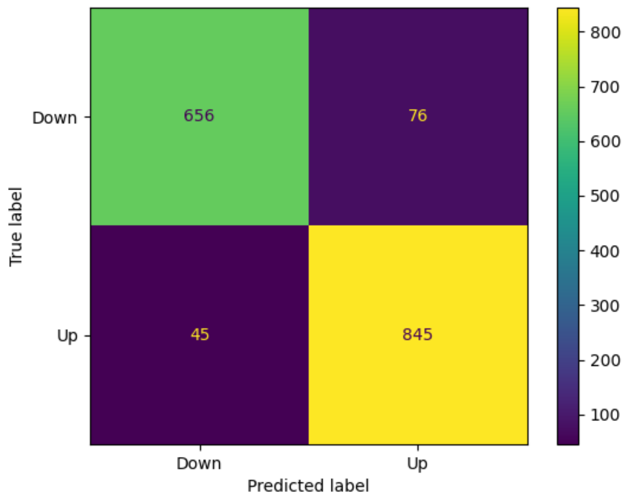

Figure 7: Logistic Regression Confusion Matrix

Fairness

Note: We were not sure about which attribute was required to be more fair as compared to the others, so it made sense for us to use "equal opportunity". Calculating the equal opportunity score is good when demographic data is present, since we do not have demographic data, so true positive rate and false positive rate was used, as seen in the code. Therefore, we have interpreted that, if our equal opportunity score is closer to 0 it means that our model has a lower bias rate and is fair.